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| <b>Caribbean Coral Reef Institute (CCRI) Water Quality Monitoring Lab</b><br><b>University of Puerto Rico – Mayagüez Campus</b><br><b>Department of Marine Sciences</b> |                 |            |  |
| Document Title                                                                                                                                                          | Document Number | Version    |                                                                                     |
| <b>SOP Dissolved Oxygen</b>                                                                                                                                             | <b>WQP 004</b>  | <b>1.0</b> |                                                                                     |

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## 1. Title

Standard Operating Procedure for Dissolved Oxygen Determination in Seawater

## 2. Applicability

This SOP applies to the determination of Dissolved Oxygen profiles at shallow (<30 m) coral reef sites around Puerto Rico.

## 3. Purpose

- 3.1. Provide laboratory-specific instructions for the CCRI Water Quality Monitoring Lab personnel to analyze dissolved oxygen profiles at shallow coral reef sites around Puerto Rico.
- 3.2. Provide water quality data to be uploaded to the EPA WQX database for use by state and federal agencies and any other parties interested in coral reef seawater dissolved oxygen data following the Quality Assurance Project Plan.

## 4. Method

- 4.1. Standard Method 4500-O/H (Baird et al., 2017).
- 4.2. Hydrolab HL4 Multiparameter Sonde is used to measure surface, bottom, and depth profiles of in situ seawater dissolved oxygen.

## 5. Health and Safety

- 5.1. Follow the safety indications for equipment and personnel suggested by the manufacturer.
- 5.2. When loosening removable parts of a manifold probe, face the probe away from your body and others, as pressure may build up within the instrument resulting in caps, sensors, or other parts that can shoot out and cause injury.

## 6. General Precautions

- 6.1 Always maintain two points of contact when deploying the HL4 sonde on the boat to avoid losing the instrument overboard.
- 6.2 Ensure the boat engines are idle and the captain approves of the sonde being deployed overboard before deploying the sonde to ensure that the instrument cables do not become entangled with the boat propeller.
- 6.3 During maintenance or replacement of the multiprobe battery, place it horizontally on a planar surface before removing the battery cap to avoid water leaks into the multiprobe battery compartment during maintenance or replacement.
- 6.4 In case water leaks into the battery compartment, remove the batteries, pour the water out, thoroughly dry the compartment, and install new batteries.
- 6.5 If there are problems with the software, call the manufacturer (800-949-3766) or email [natechsupport@otthydromet.com](mailto:natechsupport@otthydromet.com).

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- 6.6 Hydrolab HL4 Sonde requires that sensors are calibrated in the following order:
  - 6.6.1 Temperature (by manufacturer only) → conductivity → LDO → pH
- 6.7 Do not allow the sensors on the Hydrolab HL4 Sonde to dry.
  - 6.7.1 Once the sonde is returned to the boat, add clean tap water to the Hydrolab storage cup, remove the sensor guard, and replace it with the storage cup.
- 6.8 For storage after deployment:
  - 6.8.1 Rinse the handheld surveyor, the sonde sensors and the sensor guard with tap water.
  - 6.8.2 Rinse the storage cup with tap water.
  - 6.8.3 Remove the sensor guard and replace it with the storage cup while it is still wet to allow the sensors to remain humid but **not submerged**.

## 7. Training

- 7.1 Personnel shall be trained to calibrate and operate the multiparameter HL4 sonde before use.
- 7.2 Personnel shall receive training on how to complete the corresponding forms for this SOP.

## 8. Required Materials

- 8.1 Hydrolab HL4 Multiparameter Sonde
- 8.2 Hydrolab Handheld Surveyor
- 8.3 Handheld Surveyor USB cable
- 8.4 Mild detergent to clean the sensor
- 8.5 Calibration standards
- 8.6 Warm water to clean the sensor
- 8.7 Soft toothbrush to clean the sensor
- 8.8 Extra “D” battery for replacement
- 8.9 Calibration Form (Appendix I)
- 8.10 Field Form (Appendix F)

## 9. Procedure

### 9.1 Calibration

Perform a calibration check prior and after each sampling event to ensure accuracy and control for any potential sensor drift throughout the sampling event.

#### 9.1.1 Prepare the LDO calibration solution

- 9.1.1.1 Add tap water into a 1-gallon container and allow it to reach room temperature.
- 9.1.1.2 Shake the container vigorously for 40 seconds to obtain the 100% air saturated standard required for the calibration.

#### 9.1.2 Calibration Check

- 9.1.2.1 Perform the LDO sensor calibration outdoors for accurate atmospheric pressure measurements.

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- 9.1.3 Place the **HL4 sonde** and **Handheld Surveyor** on a stable/safe place to start the calibration process.
- 9.1.4 Connect **Handheld Surveyor** to **HL4 Sonde** → Press the **power button** to turn on → press the **RIGHT arrow** until you read “Surveyor HL” on the top of the display window.
  - 9.1.4.1 Select “*Real-time*” → press the **RIGHT button** until you see “BP (mmHg)” in the lower right corner of the handheld display window.
    - 9.1.4.1.1 Manually record barometric pressure shown on the display window of the **Handheld Surveyor** as you will need it later during calibration.
- 9.1.5 Press the **return button** ↵ → press the **RIGHT arrow** until you read “Surveyor HL” → select the “*Calibration*” button → select “*Hach LDO*” → press the **DOWN arrow** and select “*Check Calibration*”.
  - 9.1.5.1 Manually enter the barometric pressure you obtained on step 9.1.4.1.1 → press **OK**.
- 9.1.6 Fill the storage cup to 3/4 of its capacity with the LDO Standard Solution, attach it to the sonde and gently tap the storage cup to remove any bubbles stuck to the sensors.
  - 9.1.6.1 Make sure all sensors are submerged while sensors are facing downwards.
- 9.1.7 Write down 100.00 in the provided textbox and press the **OK button** to accept → allow for stabilization → press **OK button** again to save → select “Keep Calibration Date”.

## 9.2 Calibration

Calibration Frequency: The Hydrolab HL4 Multiparameter Sonde (OTT HydroMet) should be calibrated following manufacturers recommendation (~3 months) to ensure accuracy and control for any potential sensor drift throughout the sampling event.

### 9.2.1 LDO sensor Calibration

- 9.2.1.1 Press the **menu button** → **RIGHT arrow** until you read “Sonde\_HL4” on the top of the display window.
- 9.2.1.2 Select the “*Calibration*” button → select “*Hach LDO*” → select “*DO %SAT Calibration*”.
- 9.2.1.3 Fill the storage cup 1/3 of its capacity with the LDO standard solution, attach it to the sonde, and shake vigorously for at least 6 seconds.
  - 9.2.1.3.1 Repeat three times.
  - 9.2.1.3.2 Discard rinse water every time.
- 9.2.1.4 Fill the storage cup to 3/4 of its capacity with LDO saturated standard solution → attach the storage cup to the sonde, and make sure all sensors are submerged while facing downwards.
- 9.2.2 Write down 100.00 in the provided textbox and press the **OK button** to accept → press **OK button**.
- 9.2.3 Complete calibration check form:
  - 9.2.3.1 User name
  - 9.2.3.2 Log Notes - Include the names of all people involved in the calibration.

|                                                                                                                                                                         |                 |            |                                                                                     |
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9.2.3.3 “Calibration Interval” is left untouched – calibration will be performed prior to each sampling event.

9.2.3.4 Select “Keep Calibration Date”.

9.2.4 Record calibration results in Form Appendix I.

### 9.3 Prepare the sonde for deployment.

9.3.1 Connect the sonde to the handheld Surveyor® HL using the deployment cable and press the **power button** on the handheld.

9.3.2 After checking calibration for each sensor, attach the tap water-humid storage cup to the sonde until arrival to the sampling site.

9.3.3 Once at the sampling site, replace the storage cup with the sensor guard for deployment.

### 9.4 Sample Collection

9.4.1 While the HL4 sonde is connected to the handheld, press the **power button** → **RIGHT arrow** until you read “Sonde\_HL4”.

9.4.1.1 Select “Overview” and make sure all sensors have started properly (there should be a check mark to the right of each parameter (e.g. LDO sensor ✓)).

9.4.1.1.1 Once checked, press the **return button** ↵.

9.4.2 Without placing the HL4 sonde into the water, and while on the “Sonde\_HL4” menu, select “Real-time” → press **OK button** to start data save.

9.4.2.1 Press the **DOWN arrow button** once and then the **UP arrow button** once to select “New Log” → press **OK button**

**9.4.2.1.1** Replicate casts should be saved as “SiteCode\_R2” instead of the standard log name. **Otherwise, leave the default log name**

9.4.2.2 Select “Location” and select the corresponding “Site Code” → press **OK button**.

9.4.2.3 Press the **RIGHT arrow button** to save the settings → press **OK button** to begin data recording (you should see the number of “Measurements” on the lower left corner of the display window steadily increasing).

9.4.3 Gently lower the HL4 Sonde ~0.5 m into the water → using Sampling Log (Appendix F) record the desired parameters and their respective depths, **leaving surface temperature for last** to allow for stabilization of the sensor.

9.4.4 Lower the sonde at a slow and steady rate until it gets ~0.5m from the bottom with caution not to touch the bottom and stir sediment up.

9.4.5 Record the bottom dissolved oxygen and corresponding depth reading on the Sampling Log (Appendix F).

9.4.6 Pull the sonde back up into the boat.

9.4.7 Press **OK button** to stop recording the cast → press **return button** ↵ → turn off the handheld surveyor.

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9.4.8 If other sites are to be sampled, place the Hydrolab sonde on a bucket that holds clean tap water until arrival at the next site.

9.4.8.1 If no other sites are to be sampled, follow **step 6.8 for storage after deployment**.

## 9.5 Transfer data from handheld Surveyor to PC

9.5.1.1 Connect the surveyor to the PC using the manufacturer's USB cable and press the **power button** on the handheld.

9.5.1.2 Open the *Hydrolab Operating Software* → select the handheld surveyor from the available devices list → click on “Overview” on the navigation bar.

9.5.1.2.1 On the lower-right of the display, select the log files you wish to transfer and click “Transfer...”. → select save folder → click “Save” (these will be in Hydrolab File Format).

9.5.1.2.2 Open the Hydrolab File Format saved files → click “Export...” → click **OK** → rename the file as: *SiteCode\_Date of collection\_default log name* (e.g., BOYA20\_12MAY23\_Log\_12345\_6789) → select format type (e.g., CSV, XLSX) → click “Save”.

## 10. Documentation

10.1 Field data will be recorded in Field Form (Appendix F).

10.2 Calibration data will be recorded in Calibration Form (Appendix I).

10.3 Field and Calibration forms shall be kept in the designated file cabinet at the CCRI office.

10.4 Data contained in hard copy forms shall be digitized into the corresponding electronic spreadsheets as soon as possible after the task is completed.

10.5 All spreadsheets and depth profile logs should be uploaded to the internal CCRI Water Quality Database.

10.6 All data will be electronically archived and backed up online.

## 11. Cleaning and Maintenance

11.1. After each sampling event, rinse the handheld surveyor, the sensors, the storage cup, and the sensor guard thoroughly with clean warm fresh water.

11.2. After each sampling event, inspect sensors carefully for physical damage.

11.3. In case of unwanted material deposits over the sensors, gently clean sensors with a soft brush and mild detergent. Do not use dishwashing liquid or detergents with skin moisturizer as these can create films over the probes.

11.4. For long-term storage (>24hr), rinse the storage cup and sensors with tap water and screw the storage cup on the sensors while it is still humid. **No sensor should be permanently touching or be submerged in water.**

11.5. Do not fill the storage cup with distilled water as this may damage the probes.

11.6. Replace sonde “D” battery as necessary.

11.7. The handheld surveyor “Overview” shows when the battery charge is low.

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- 11.8. Perform a monthly preventive maintenance
- 11.9. Carefully inspect sensors, O-rings, connective pins, and physical integrity of the sonde in general.
  - 11.9.1. Refer to HL Series Sensors DOC026.97.80349 document for instructions.
  - 11.9.2. Lubricate or replace O-rings, if necessary.
- 11.10. Follow manufacturer indications for sensor maintenance.
  - 11.10.1. Refer to “HL Series Sonde DOC026.97.80348” document for instructions.

## 12. Troubleshooting

- 12.1 In case of reading errors, check the sensor path for bubbles, grease, or other contamination.
- 12.2 In case of unstable readings, contact the manufacturer to replace the sensor.

## 13. QA/QC

- 13.1 As per QAPP Table 4, seawater Dissolved Oxygen should be measured in milligrams per liter (mg/L) with an accuracy of  $\pm 0.2$  mg/L ( $\pm 20\%$ ) and a precision of  $< 20\%$ .
- 13.2 In accordance with the QAPP, 10% of all Hydrolab casts will be repeated to ensure precision of replicate surface and bottom measurements is  $< 20\%$ .
- 13.3 All field samples should adhere to typical seasonal dissolved oxygen values observed in coastal waters of Puerto Rico.
- 13.4 Any seemingly erroneous measurements will be flagged for further review in accordance with the QAPP.

## 14. References

- 14.1. Baird, R. B., Eaton, A. D., & Rice, E. W. (2017). Standard methods for the examination of water and wastewater (Vol. 23). E. W. Rice (Ed.). Washington, DC: American Public Health Association
- 14.2. Hydrolab (2014). Hydrolab Operating Software. DOC026.53.80346I
- 14.3. Hydrolab (2017). HL Series Sonde User Manual. DOC026.97.80348
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